Course notes

The co-ordinate calculations of this chapter are done in an orthonormal space coordinate system.

Scalar product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos\theta$ where: $\theta \in [0, \pi]$ is their angle. If one is zero, the scalar product is 0.

$$\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal coordinate system, $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The scalar product is symmetrical and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their scalar product is zero.

Example. $\vec{u} = (1,2,2)$ and $\vec{v} = (2,-1,0)$ are orthogonal: $2 \quad -2+0=0$. Their norms are 3 and $\overline{5}$.

Straight, plan and normal vector

Two lines are orthogonal if their guiding vectors are. A right is orthogonal to a plane if it is orthogonal to two non-collinear directions of this plane. A normal vector to one plane is a non-zero orthogonal vector in all its directions.

Why two directions are enough. If $\vec{n}\cdot\vec{u} = \vec{n}\cdot\vec{v} = 0$, then, for everything $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n}\cdot\vec{w} = s\vec{n}\cdot\vec{u} + t\vec{n}\cdot\vec{v} = 0$.

Vector condition of belonging

If A belongs to the plan $\mathcal{P}$ of normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM}\cdot\vec{n} = 0$.

Orthogonal projection and distance

The orthogonal project H of M on a plan is the only point of the plan such as $\overrightarrow{MH}$ or normal in the plan ; if M belongs to the plan, $H = M$. The distance from M on plan east MH. On a director's right $\vec{u}$, the projected verifies $H \in d$ and $\overrightarrow{MH}\cdot\vec{u} = 0$.

Demonstration of minimum. For any point P of the plan, $\overrightarrow{MH} \perp \overrightarrow{HP}$, therefore $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality takes place only for $P = H$. The project is the point of the nearest plan.

Method on a right. If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and resolve $(A - M + t\vec{u})\cdot\vec{u} = 0$. Thus $t = \frac{\overrightarrow{AM}\cdot\vec{u}}{\|\vec{u}\|^2}$.

Angles, areas and volumes

For an angle $\widehat{ABC}$use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos\widehat{ABC} = \frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{BA\times BC}$. For the acute angle between two straight lines, take the absolute value of the quotient. The area of a triangle is $\frac{bh}{2}$; the volume of a tetrahedron is $\frac{\mathcal{A}_{\text{base}}\,h}{3}$, where h is an orthogonal distance. Units are squared and cubed respectively.

Geometrical developments and locations

By bilinearity,

$$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u}-\vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2.$$

It is therefore also possible to calculate $\vec{u}\cdot\vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u}-\vec{v}\|^2)$. For two separate points A, B of midpoint I, the locus of the points M such that $MA = MB$ is the plan passing through I normal $\overrightarrow{AB}$: the *plan* Ombudsman. Indeed, by writing vectors since I, developments give $MA^2 - MB^2 = 2\overrightarrow{IM}\cdot\overrightarrow{AB}$. Equality of distances is therefore equivalent to $\overrightarrow{IM}\cdot\overrightarrow{AB} = 0$. **Supplement, without dedicated evaluation in 2026·27.** Two planes are perpendicular when their normal vectors are orthogonal. Do not confuse this property with the parallelism of the plans, which corresponds to normals Collinear.

See a projection, then calculate it

11.1 A scalar product may be negative

First use: S48.

Definition · Scalar product

For two non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos\theta$ where: $\theta \in [0; \pi]$ is their angle. orthonormal coordinate system, this product is $xx' + yy' + zz'$. Invalidity characterizes the orthogonality of directions.

Solved Example · Recognize an obtuse angle

For $\vec{u} = (1;0;0)$ and $\vec{v} = (-1;1;0)$ the product is -1, norms are 1 and $\sqrt{2}$, therefore $\cos\theta = -1/\sqrt{2}$ and $\theta = 3\pi/4$. A negative product indicates an obtuse or flat angle, not a negative length.

A1 · Immediate application Calculate the angle between $(1;1;0)$ and $(1;0;1)$ in an orthonormal coordinate system. Could we use the same calculation in any database?

11.2. Two directions are required

First use: S51.

Property · Orthogonality Criterion to a Plan

A non-zero vector is normal at a plane if it is orthogonal with two non-directive vectors
It is then orthogonal to any combination of these two.
vectors, therefore in any direction of the plan.

Example resolved · Test the three conditions

For $\vec{u} = (1;1;0)$, $\vec{v} = (0;1;1)$ and $\vec{n} = (1;-1;1)$, $\vec{n} \cdot \vec{u} = 0$ and $\vec{n} \cdot \vec{v} = 0$. The two directions do not are non-colonial and $\vec{n} \neq 0$, therefore $\vec{n}$ is normal to the plan they lead.

A2 · Counterexample "Being orthogonal to one right of the plane is enough to be normal on the plane." Refute with two perpendicular directions of the horizontal plane. Also explain the difference between orthogonal and perpendicular straight lines.

Example resolved · A locus of equidistant points

Let A=(-1;0;0) and B=(1;0;0). Their midpoint is O. For M=(x;y;z), $MA^2 - MB^2 = (x+1)^2 - (x-1)^2 = 4x$. Thus $MA = MB$ is equal to x=0: the locus is the plan passing through O, normal to $\overrightarrow{AB}$. It is not limited to the mediator drawn in one face.

A3 · Application and justification For A=(0;0;0), B=(2;2;0), describe geometrically the location of the Equidistant points of A and B. Use distance squares to give a relationship between coordinates; find two points of this locus.

11.3 The foot is the nearest point

First use: S54 and S55.

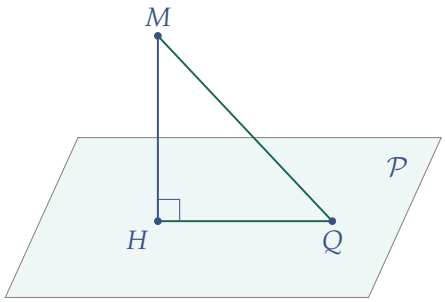


Figure 11.1. Coding represents $MH \perp \mathcal{P}$. For each Q of the plan, $MQ^2 = MH^2 + HQ^2$.

Demonstration. If H is the orthogonal project of M on a plan $\mathcal{P}$, for everything $Q \in \mathcal{P}$, the triangle MHQ is right-angled at H. Thus $MQ^2 = MH^2 + HQ^2 \geq MH^2$. Equality requires $HQ = 0$, therefore $Q = H$. The minimum is reached in a single point. The reasoning is the same for a straight line. □

Solved Example · Projecting on a straight line without plane equation

Let $A = (1;0;0)$, $M = (0;2;3)$ and a right of point A and direction $\vec{u} = (1;1;0)$. Write $\overrightarrow{AH} = t\vec{u}$ gives $H = (1+t;t;0)$. The condition $\overrightarrow{MH} \cdot \vec{u} = 0$ becomes $(1+t) + (t-2) = 0$, therefore $t = 1/2$. Thus $H = (3/2;1/2;0)$ and $MH^2 = 27/2$. Check both membership to the right and orthogonality.

A4 · Partial guidance Project $M = (2;3;1)$ on the right passing through O and direction $(1;1;0)$. Determine project and distance.

A5 · Choice of strategy Project $M = (1;2;3)$ on the plan generated by the $\vec{i}, \vec{j}$. To justify geometrically the foot coordinates, then demonstrate that any other point of the plane is further away.

11.4. The same size by two calculations

First use: S56.

Example Solved · Find a Height

In an orthonormal coordinate system, $O = (0;0;0)$, $A = (6;0;0)$, $B = (0;4;0)$, $C = (0;0;3)$. The triangle OAB is rectangle of area 12 and OC is perpendicular to its plane; the volume is 12 $\times 3/3 = 12$. If the ABC area is 3 $\overline{29}$, its height from O is 3 $V/\mathcal{A}(ABC) = 12/\ \overline{29}$.

A6 · Summary In this tetrahedron, calculate AB, AC and $\overrightarrow{AB} \cdot \overrightarrow{AC}$. Deduce the cosine and then the sine of angle $\widehat{BAC}$, check ABC's area and find the height from O.

A first step: Use $\mathcal{A}(ABC) = \frac{1}{2}$AB AC sin $\widehat{BAC}$. The angle of a triangle has a positive sine.

A7 · Repair an audit A candidate gives a point H of the plan and affirms "it is the project since H What second audit is missing? Give two points in the plan. $z = 0$ to show that belonging alone is not enough.

Wait a minute at the courthouse. Describe the projection figure by naming the outer point, the plane, the foot, and right angle; explain why distance is not simply calculated at any point of the plan.

Review postponed · S70. Recalculate foot with parametric representation and equation Cartesian; the two geometric conditions remain identical.

Research activity

Nearest straight point · 35 min.

In an orthonormal coordinate system of space, we consider $A(1;0;0)$, $M(0;2;3)$ and the right d passing through A, of a policy vector $\vec{u} = (1;1;0)$. We're looking for the point of d closest to M.

1. Express, based on a real parameter t, the coordinates of a point H of d.
2. Determine the value of t for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the point H and so obtained.
3. Calculate distance MH.
4. Let P any point of d. Demonstrate that $MP^2 = MH^2 + HP^2$, and then deduce that H is the only point of d closest to M.